Elementary particle physics and group theory

In the 1950s physicists found so many new particles that Fermi is said to have remarked that if he could remember all their names he would have become a botanist. Order came from an unexpected quarter: group theory, the mathematics of symmetry. It predicted new particles, led to the quarks and finally became the very language of the Standard Model, in which the forces of nature follow from symmetries. This page explains how – with two interactive figures.

The particle zoo

In 1932 the world looked simple: atoms consisted of protons, neutrons and electrons, and light of photons. But that same year Carl Anderson found the positron, which Dirac had predicted. Then came the muon (1936), the pion (1947), and with the new accelerators of the 1950s a stream of short-lived particles: K mesons, Λ, Σ, Ξ, Δ and many more. Some of them behaved "strangely": they were produced quickly in strong interactions but decayed slowly. In 1953 Gell-Mann and Kazuhiko Nishijima explained this with a new conserved quantum number, *strangeness*.

The question was whether there was a pattern behind it all. The answer came from a branch of mathematics that most physicists of the time knew only superficially.

What is a group?

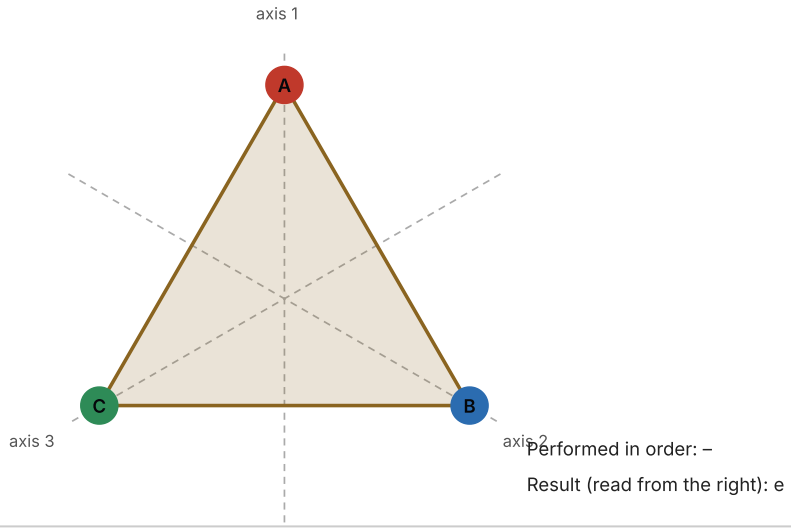
A symmetry is an operation that leaves something unchanged. An equilateral triangle can be rotated by 120° or 240°, or reflected in three axes, and it looks the same as before. Together with "do nothing" there are six symmetries. Performing two symmetries one after the other gives another symmetry. Mathematically this is a *group*: a set with a composition that obeys four rules.

The group axioms

1. **Closure:** $a \cdot b$ is again in the group.
2. **Associativity:** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
3. **Identity:** there is an e with $e \cdot a = a \cdot e = a$.
4. **Inverses:** for each a there is an a^{-1} with $a \cdot a^{-1} = e$.

Do they commute?

The rules do not require $a \cdot b = b \cdot a$. For the symmetries of the triangle it is often false: rotate then reflect is not the same as reflect then rotate. Groups in which the order does not matter are called *abelian* (after Niels Henrik Abel). Electromagnetism is built on an abelian group, the nuclear forces on non-abelian ones.



The group table (Cayley table) for the symmetry group of the triangle, called D_3. Here row a and column b give the product $a \cdot b$, i.e. "first b, then a":

·	e	r	r^2	s_1	s_2	s_3
e	e	r	r^2	s_1	s_2	s_3
r	r	r^2	e	s_3	s_1	s_2
r^2	r^2	e	r	s_2	s_3	s_1
s_1	s_1	s_2	s_3	e	r	r^2
s_2	s_2	s_3	s_1	r^2	e	r
s_3	s_3	s_1	s_2	r	r^2	e

Continuous groups

The triangle's group has six elements. A circle, by contrast, can be rotated through any angle, and rotations in space form an infinite, connected group, $SO(3)$. Such groups are called Lie groups after the Norwegian mathematician Sophus Lie. The crucial point is that the whole group can be understood from rotations through an *infinitesimal* angle. These are described by three *generators*, and all the information about the group lies in what happens when they are interchanged:

$$[J_x, J_y] = J_x J_y - J_y J_x = i\hbar J_z \quad \text{(and cyclically)}$$

These are exactly the rules of quantum mechanics for angular momentum. The electron's spin belongs to the group $SU(2)$ – the 2×2 unitary matrices with determinant 1 – which is almost the same as $SO(3)$, except that a rotation through 360° puts a minus sign in front of the wave function. Only after 720° is one back where one started. This is no mathematical nicety: it was measured with neutrons in 1975. See also Rotation in space and the quaternions, which are $SU(2)$ in disguise.

A *representation* is a way in which the group can act on concrete states. For $SU(2)$ there is one for each spin $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$, and it has $2j+1$ states – which is why a spin ½ has two settings and a spin 1 three. In particle physics a representation corresponds to a family of particles, a *multiplet*, linked by the symmetry.

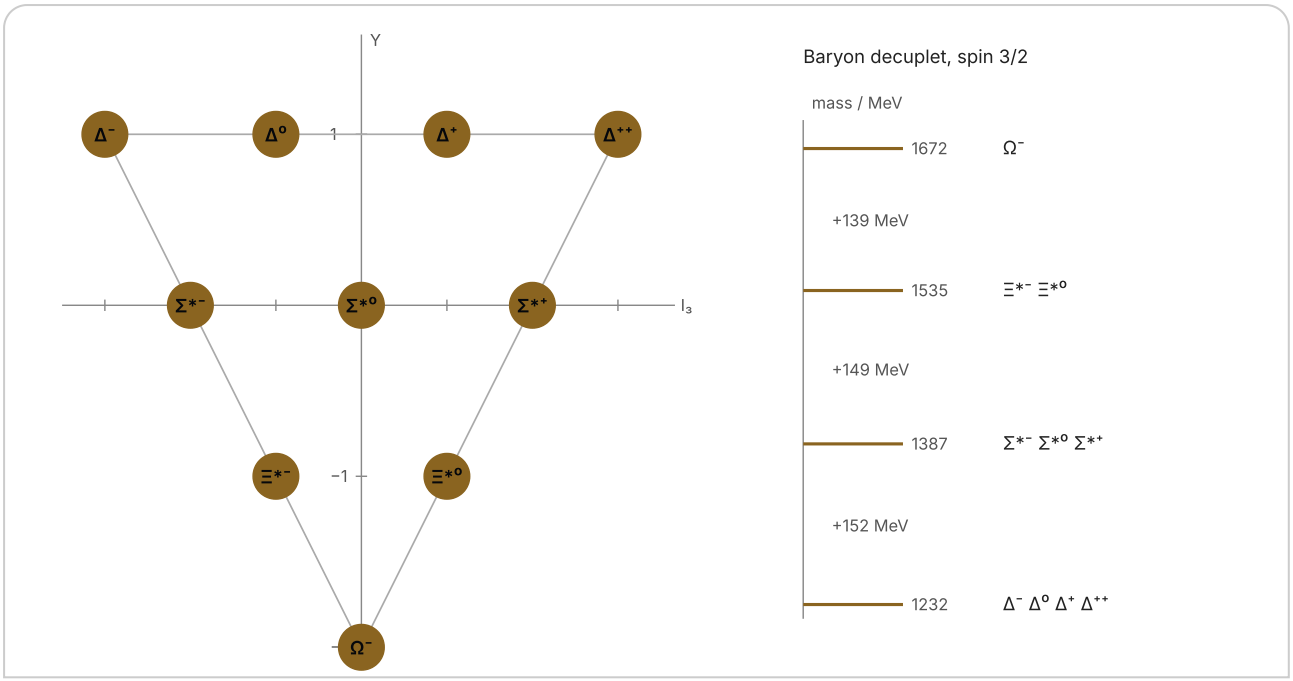
Isospin – the proton and the neutron as one particle

The proton and the neutron have almost the same mass (938.3 and 939.6 MeV), and the nuclear force treats them almost alike. In 1932 Heisenberg proposed regarding them as two states of a single particle, the nucleon, just like spin up and spin down. The abstract symmetry is called isospin and is an $SU(2)$ symmetry: the proton has $I_3 = +\frac{1}{2}$, the neutron $I_3 = -\frac{1}{2}$. The pions π^+, π^0, π^- form a triplet with isospin 1. That their charges differ is due to electromagnetism, which breaks the symmetry slightly – and explains the small difference in mass.

SU(3) and the Eightfold Way

With strangeness the particles had two quantum numbers besides spin and parity: I_3 and the hypercharge Y (= baryon number + strangeness). Plot particles of the same spin in a diagram with I_3 horizontally and Y vertically, and regular hexagons and triangles appear. In 1961 Murray Gell-Mann, and independently Yuval Ne'eman, realised that the patterns are representations of the group $SU(3)$. Gell-Mann called the idea *the Eightfold Way* after the Buddha's eight steps, because the lightest baryons and mesons come in groups of eight.

But $SU(3)$ also has a representation with ten members. In 1962 nine heavy baryons with spin 3/2 were known, arranged in a triangle with an empty apex. Gell-Mann predicted the missing particle: it should have charge −1, strangeness −3 and a mass of about 1680 MeV, because the masses in the triangle rise in almost equal steps. In 1964 the Ω^- was found at Brookhaven in a bubble-chamber photograph – with a mass of 1672 MeV. It was group theory's Mendeleev moment.



Quarks and colour

Why these particular patterns? In 1964 Gell-Mann, and independently George Zweig, proposed that all the representations can be built from the simplest one, the triplet: three *quarks*, u, d and s, with charges $+\frac{2}{3}$, $-\frac{1}{3}$ and $-\frac{1}{3}$. Gell-Mann took the name from a line in James Joyce's *Finnegans Wake*. Mesons consist of a quark and an antiquark, baryons of three quarks, and the rules of group theory for combining representations give exactly the observed multiplets:

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}, \qquad \mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1}$$

There was a problem. The Δ^{++} consists of three u quarks with parallel spins in the same state – exactly what Pauli's exclusion principle forbids. The solution (Greenberg 1964, Han and Nambu 1965) was that each quark has an extra quantum number with three values, called *colour*: red, green and blue. The three quarks in the Δ^{++} each have a different colour. All the particles we observe are colourless, and quarks never appear on their own. In 1968–69 electron-scattering experiments at SLAC showed that the proton really does have point-like constituents. The charm quark followed in 1974, bottom in 1977 and top in 1995.

Forces from symmetry

The deepest application of group theory is the *gauge principle*. The wave function of quantum mechanics can be multiplied by a phase $e^{i\alpha}$ without changing anything measurable – a $U(1)$ symmetry. If one demands that the phase can be chosen differently at each point in space and time (a *local* symmetry), the equations only work if there is a field that couples to the charge and

compensates for the phase differences. That field is electromagnetism, and its quantum is the photon.

In 1954 Chen Ning Yang and Robert Mills extended the idea to non-abelian groups such as $SU(2)$. The number of force-carrying particles equals the dimension of the group, and because the generators do not commute, the force carriers interact with each other.

Force	Group	Force carriers	Number = dimension of the group
Electromagnetism	$U(1)$	photon	1
Electroweak	$SU(2) \times U(1)$	W^+, W^-, Z^0, photon	3 + 1
Strong (quantum chromodynamics)	$SU(3)_{\text{colour}}$	gluons	8

In a pure gauge theory the force carriers are massless, but the W and Z are heavy. The solution was the Higgs mechanism (Englert and Brout; Higgs; Guralnik, Hagen and Kibble, 1964): a field filling all of space breaks the symmetry spontaneously and gives the particles mass. Sheldon Glashow (1961), Steven Weinberg (1967) and Abdus Salam (1968) united the weak and electromagnetic forces in a single theory. The W and Z were found at CERN in 1983, and the Higgs boson in 2012.

In quantum chromodynamics the gluons themselves carry colour. This gives a force that becomes weaker at short distances (*asymptotic freedom*, Gross, Wilczek and Politzer 1973) and stronger at large ones – which is why quarks cannot be pulled free.

The Standard Model

The Standard Model has the combined symmetry group $SU(3) \times SU(2) \times U(1)$. Matter consists of three generations of quarks and leptons that differ only in mass:

	1st generation	2nd generation	3rd generation	Force carriers
Quarks	u up · +⅔ · 2.2 MeV	c charm · +⅔ · 1.27 GeV	t top · +⅔ · 173 GeV	g gluon · 0 · 0
	d down · −⅓ · 4.7 MeV	s strange · −⅓ · 93 MeV	b bottom · −⅓ · 4.18 GeV	γ photon · 0 · 0
Leptons	e electron · −1 · 0.511 MeV	μ muon · −1 · 106 MeV	τ tau · −1 · 1.78 GeV	Z Z boson · 0 · 91.2 GeV
	ν_e electron neutrino · 0 · < 1 eV	ν_μ muon neutrino · 0 · < 1 eV	ν_τ tau neutrino · 0 · < 1 eV	$W^\pm$ W boson · ±1 · 80.4 GeV
	H Higgs boson · spin 0 · 125 GeV			

Name · charge (in units of e) · mass. The masses of the light quarks are model-dependent. Every particle has an antiparticle, and every quark comes in three colours.

Broken symmetries

Just as important as the symmetries is where they are broken. In 1956 Lee and Yang proposed that the weak force does not respect mirror symmetry (parity, P), and Chien-Shiung Wu demonstrated it in 1957. In 1964 James Cronin and Val Fitch found that even the combination of mirroring and replacing particles by antiparticles (CP) is slightly broken in the decay of K mesons. CP violation is one of the preconditions for the universe consisting of matter rather than equal amounts of matter and antimatter. The combination CPT (with time reversal), by contrast, follows from relativity and quantum mechanics and has never been seen to be broken.

Emmy Noether's theorem links every continuous symmetry to a conservation law: symmetry in time to energy, symmetry in space to momentum, $U(1)$ phase symmetry to electric charge. Spontaneously broken symmetries are everywhere in physics: in a magnet, in a superconductor and in the Higgs field.

What is missing?

The Standard Model is the most precisely tested theory in the history of science, but it cannot be the whole story:

- **Neutrino masses.** Neutrinos oscillate between types (Super-Kamiokande 1998, SNO 2001), so they have mass. The model must be extended to explain this.
- **Dark matter.** About 85 % of the matter in the universe is not described by the model.

- **Grand unified theories.** In 1974 Georgi and Glashow proposed that the three groups are parts of a single larger one, $SU(5)$. This predicts that the proton decays, but its lifetime has been measured to exceed about 10^{34} years – and the simplest version is ruled out.
- **Gravity.** The Standard Model does not include gravitation. See String theory – and its critics.
- **Why?** The model has about 19 free parameters (plus those of the neutrinos), which must be measured. Why are there three generations, and why do the particles have the masses they have?

Further reading

- David Griffiths: *Introduction to Elementary Particles* (2nd ed., Wiley 2008)
- Anthony Zee: *Group Theory in a Nutshell for Physicists* (Princeton 2016)
- Howard Georgi: *Lie Algebras in Particle Physics* (2nd ed., 1999)
- Frank Close: *The Infinity Puzzle* (2011)
- George Johnson: *Strange Beauty. Murray Gell-Mann and the Revolution in Twentieth-Century Physics* (1999)
- The biographies of Gell-Mann, Noether, Pauli, Dirac, Wu and Feynman

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